1. recall that many chemicals in living things are natural polymers (limited to carbohydrates and proteins);
2. recall that cellulose, starch and sugars are carbohydrates which consist of carbon, hydrogen and oxygen;
3. recall that amino acids and proteins consist mainly of carbon, hydrogen, oxygen and nitrogen;
4. understand that there is continual cycling of elements through consumption of living organisms and decay;
5. **describe the main stages of the nitrogen cycle;**
6. understand that where crops are harvested, elements such as nitrogen, **potassium and phosphorus**, are lost from the soil so that the land becomes less fertile unless these elements are replaced;
7. recall and explain the methods used by organic and intensive farmers to maintain the fertility of soils used to grow crops;
8. understand that yields from crops my be reduced by pests and disease;
9. understand that organic and intensive farmers use different methods to protect crops against pests and diseases, and that these can have different effects on the environment;
10. understand that farmers have to follow the UK national standards if they want to claim that their products are organic;
11. when provided with information about the methods used in farming:
    - can identify the groups affected and the main benefits and costs of a course of action for each group;
    - can explain the idea of sustainable development, and apply it to specific situations;
    - show awareness that scientific research and applications are subject to official regulations and laws;
    - **can distinguish between what can be done (technical feasibility) from what should be done (values);**
    - **can explain why different courses of action may be taken in different social and economic contexts.**